

JAVA COLLECTIONS

Program: Collection Methods Demonstration

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Aim

To demonstrate commonly used methods of the Java Collection interface such as add(), addAll(), contains(), containsAll(), size(), isEmpty(), remove(), removeAll(), iterator(), and clear().

Requirements

Hardware: Computer/Laptop with keyboard and mouse.

Software: JDK 8 or later, Java compiler, and any Java IDE or Command Prompt/Terminal.

Algorithm / Steps

1. Create a Collection of String objects using ArrayList.
2. Add three student names using add().
3. Create another collection using Arrays.asList() and add it using addAll().
4. Check whether a particular student exists using contains().
5. Check whether all elements of another collection exist using containsAll().
6. Display the collection size using size().
7. Check whether the collection is empty using isEmpty().
8. Remove one student using remove().
9. Remove all elements of the second collection using removeAll().
10. Create an Iterator and traverse the remaining elements using hasNext() and next().
11. Clear the collection using clear().
12. Check isEmpty() again and display the final result.

UML / Class Diagram

```
CollectionMethodsDemo
-----
+ main(String[] args) : void
-----
uses
|
v
Collection<String>
|
+-- ArrayList<String>
+-- Iterator<String>
+-- Arrays.asList()
```

Source Code

```
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Collection;
import java.util.Iterator;

class CollectionMethodsDemo {

    public static void main(String[] args) {

        // Creating a Collection
        Collection<String> students = new ArrayList<>();

        // add(E e)
        students.add("Rahul");
        students.add("Priya");
        students.add("Arjun");

        System.out.println("After add(): " + students);

        // addAll(Collection c)
        Collection<String> newStudents =
            Arrays.asList("Anu", "Kiran");

        students.addAll(newStudents);

        System.out.println("After addAll(): " + students);

        // contains(Object o)
        System.out.println("Contains Rahul? "
            + students.contains("Rahul"));

        // containsAll(Collection c)
        System.out.println("Contains Anu and Kiran? "
            + students.containsAll(newStudents));

        // size()
        System.out.println("Size: " + students.size());

        // isEmpty()
        System.out.println("Is collection empty? "
            + students.isEmpty());

        // remove(Object o)
        students.remove("Arjun");

        System.out.println("After remove(): " + students);

        // removeAll(Collection c)
        students.removeAll(newStudents);

        System.out.println("After removeAll(): " + students);

        // iterator()
        System.out.println("Using iterator():");

        Iterator<String> iterator = students.iterator();

        while (iterator.hasNext()) {
            System.out.println(iterator.next());
        }

        // clear()
        students.clear();

        System.out.println("After clear(): " + students);

        // isEmpty() after clear
        System.out.println("Is collection empty now? "
            + students.isEmpty());
    }
}
```

Compilation / Execution Commands

Save the program as **CollectionMethodsDemo.java**.

```
javac CollectionMethodsDemo.java
java CollectionMethodsDemo
```

The first command compiles the Java source file and the second command executes the program.

Output

```
After add(): [Rahul, Priya, Arjun]
After addAll(): [Rahul, Priya, Arjun, Anu, Kiran]
Contains Rahul? true
Contains Anu and Kiran? true
Size: 5
Is collection empty? false
After remove(): [Rahul, Priya, Anu, Kiran]
After removeAll(): [Rahul, Priya]
Using iterator():
Rahul
Priya
After clear(): []
Is collection empty now? true
```

Result

The CollectionMethodsDemo program was successfully implemented and executed. The program demonstrates the specified Collection methods and iterator traversal, and the final collection is empty after clear().